

MECH 191 — Metallurgy

Semester Alloy Research Portfolio

Fall 2026 • Leeward Community College • Instructor: Kai Santos

Student Name: _____

Student ID: _____

Semester / Year: _____ Section: _____

How to Use This Portfolio

This portfolio contains 15 modules — one per lecture week of MECH 191. Each module assigns one or more alloys from the course specimen set and asks you to research their real, published composition, properties, processing, and applications, then apply that research to the week's concepts through short-answer and critical-thinking questions. The physical specimen kit is used for in-class demonstrations throughout the semester, but this portfolio itself is a research deliverable — no microscope work or lab session is required to complete it.

For each module you will: (1) research the assigned alloy(s) using credible technical sources, (2) record what you find with a proper citation, (3) answer the short-answer questions connecting your research to that week's specimens, and (4) answer the critical-thinking questions applying that week's engineering equation or concept.

Completed modules are kept together and submitted as a single portfolio at the end of Week 16.

Portfolio Rules

Source citation. Every module must record where each piece of data came from — publication/database name, edition or URL, and the specific value or table referenced. An unsourced number earns no credit.

Closest available data. If a published value isn't available for the exact grade/temper/condition specified, use the closest documented equivalent (a related grade, a typical/nominal value, a similar temper) and note the substitution in your answer — that's expected, not an error.

Missed sessions. Because this is a research deliverable rather than a live lab exercise, a module can be completed independently if a class session is missed — check with the instructor on the deadline.

Research Sources Used Throughout This Portfolio

Computer or internet access, needed for every module to research alloy properties and locate citable sources. Not repeated in each module's own Research Sources list below. Good general-purpose sources across every module: MatWeb (matweb.com) for composition and mechanical-property datasheets, AZoM (azom.com) for alloy overview and property articles, and manufacturer or mill datasheets (most producers post these as freely downloadable PDFs, no login required). For aluminum grades, the Aluminum Association's free composition-limits reference ("Pink Sheets," aluminum.org) is a good cross-check. For copper and brass grades, the Copper Development Association's alloy pages (alloys.copper.org) are free and detailed. This course's own reference pages (AISI/SAE Steel Codes, Aluminum Alloy & Temper Designations, Copper/Brass/Bronze Codes, Titanium Grades, Hardness Testing Scales, PREN, and Quality Standards) are also valid starting points and are called out by module below where most relevant. Every source named in this portfolio is free to access and requires no account or purchase — the official ASM Handbook and ASTM standards (E112, E45, etc.) are NOT required or expected; they sit behind institutional/library subscriptions or purchase, so they are intentionally left out of this portfolio's source list.

Conducting Your Research (Applies to Every Module)

Confirm you're citing a real published value tied to the specific alloy and condition/temper named in the module — not a generic "typical steel" number. Record the source now, before you start writing your answers: publication/database name, edition/URL, and the table or section referenced. Where a range is published (e.g., "70–95 HV"), record the range and note which end you're using for calculations. Cross-check a surprising number against a second source before using it in a calculation — published data varies more than you'd expect between sources, and that variation is itself worth noticing.

MODULE 1 · WEEK 1 | Atomic Structure & Bonding

Grain boundaries are the 2-D interfaces between adjacent crystals. Their size, shape, and misorientation angle directly control strength, toughness, and ductility — the Hall-Petch relationship states that smaller grains produce stronger metals.

Alloy of Record: A — 1018 Low-Carbon Steel (normalized)

Research Sources: AISI/SAE Steel Designation Codes (course reference page), AZoM.com or MatWeb entry for 1018, a mill certificate or manufacturer datasheet

Step 1 — Research & Documentation

Record for 1018 steel, normalized condition: composition (wt% C, Mn, and other key elements); mechanical properties (yield strength, tensile strength, elongation, hardness — cite your source); the heat-treatment route that produces "normalized"; typical applications and why this alloy/condition suits them; a typical ASTM E112 grain size number if you can find one published for this grade and condition.

Step 2 — Short-Answer Questions

Q1. Describe the expected grain structure of normalized 1018 (grain shape — equiaxed or elongated — and approximate size) and explain how the normalizing process produces it.

Answer:

Q2. Atomic bonding in steel is metallic. What property of metallic bonding allows grain boundaries to exist without the material cracking?

Answer:

Q3. Using a published ASTM E112 grain-size number (or a typical range) for normalized 1018, estimate the average grain diameter.

Answer:

Q4. What role do MnS inclusions play in this steel's mechanical behavior (anisotropy, machinability, ductility)?

Answer:

Step 3 — Engineering Equations & Concept Check

Hall-Petch Relationship

$$\sigma_y = \sigma_0 + k_y / \sqrt{d}$$

σ_y — yield strength · σ_0 — friction stress (intrinsic lattice resistance to dislocation motion) · k_y — Hall-Petch (strengthening) coefficient, material-specific · d — average grain diameter

C1. Two identical steel samples have the same composition and heat treatment, but Sample X has a finer grain structure than Sample Y. Using the Hall-Petch relationship, explain which sample you expect to have the higher yield strength and why, in terms of dislocation motion.

Answer:

C2. The Hall-Petch relationship predicts that yield strength increases without limit as grain size shrinks toward zero. Is this physically realistic? What do you think eventually limits how much refining the grain size can strengthen a metal?

Answer:

MODULE 2 · WEEK 2 | Mechanical Properties

Mechanical properties emerge from microstructure. Pearlite is harder and stronger than ferrite; a finer grain size raises yield strength (Hall-Petch). This module connects what you research about a material's structure to what a test lab measures.

Alloys of Record: A — 1018 steel (normalized) · B — 1045 steel (normalized) — reference HV value used for Q2 (Appendix A)

Research Sources: AISI/SAE Steel Designation Codes, AZoM.com, MatWeb entries for 1018 and 1045

Step 1 — Research & Documentation

For both 1018 and 1045 (normalized): composition (wt% C especially), published hardness (HV or convertible scale — cite source), published tensile/yield strength, and expected ferrite/pearlite ratio based on carbon content.

Step 2 — Short-Answer Questions

Q1. Compare Specimens A (1018) and B (1045). Which has more pearlite, and why, given 1045's higher carbon content?

Answer:

Q2. Specimen B (1045 steel, normalized) has a published/typical hardness of about 170 HV (Appendix A). Using $HV = 1.854 \times F / d_{avg}^2$ with $F = 1 \text{ kgf}$, solve for d_{avg} — the diagonal length a real Vickers indent would have produced at that reading. Show your algebra.

Answer:

Q3. Hall-Petch: $\sigma_y = \sigma_0 + k/\sqrt{d}$. If grain size is refined by 50%, what happens to yield strength? (Qualitative answer is fine.)

Answer:

Q4. Which specimen would you expect to have higher tensile strength, and why? Does the expected ferrite/pearlite ratio support that prediction?

Answer:

Step 3 — Engineering Equations & Concept Check

Engineering Stress, Strain & Hooke's Law

$$\sigma = F / A_0 \quad \epsilon = \Delta L / L_0 \quad \sigma = E \cdot \epsilon \text{ (elastic region only)}$$

σ — engineering stress; F — applied axial load; A_0 — original cross-sectional area · ϵ — engineering strain; ΔL — change in gauge length; L_0 — original gauge length · E — elastic (Young's) modulus

C1. Two tensile specimens are pulled to the same engineering stress — one is pearlitic, one is ferritic. Which one do you expect to show more strain (elongation) at that stress level, and why does the microstructure — not just the stress — determine the answer?

Answer:

C2. Hooke's law ($\sigma = E\epsilon$) only holds in the elastic region. In words, what physically happens inside the metal's crystal structure once loading passes the yield point that makes stress and strain no longer proportional?

Answer:

MODULE 3 · WEEK 4 | Carbon & Alloy Steels

The iron-carbon phase diagram predicts the equilibrium ferrite and pearlite fractions at a given carbon content. Comparing three normalized steels spanning the practical carbon range — low-carbon 1018, medium-carbon 1045, and high-carbon 1075 — shows how composition alone controls the ferrite/pearlite balance, and therefore hardness and strength, without any heat treatment involved.

Alloys of Record: A — 1018 steel (normalized) · B — 1045 steel (normalized) · C — 1075 steel (normalized)

Research Sources: AISI/SAE Steel Designation Codes, AZoM.com or MatWeb entries for 1018/1045/1075, a freely available iron-carbon phase diagram (e.g., a materials-science course page or Wikipedia)

Step 1 — Research & Documentation

Composition (wt% C) for 1018, 1045, and 1075; published hardness for each (Appendix A has reference values for C); the eutectoid composition and its significance on the iron-carbon diagram.

Step 2 — Short-Answer Questions

Q1. Draw a simple schematic of the iron-carbon phase diagram and mark where 1018, 1045, and 1075 each fall at room temperature after normalizing.

Answer:

Q2. Using the lever rule, calculate the pearlite fraction for 1018 (~0.18% C), 1045 (~0.45% C), and 1075 (~0.75% C). Rank the three by pearlite fraction.

Answer:

Q3. Pearlite lamellar spacing is controlled primarily by cooling rate, not carbon content. Explain why increasing carbon content (1018 → 1045 → 1075) increases pearlite fraction but does not, by itself, change lamellar spacing.

Answer:

Q4. 1075 has roughly four times the carbon content of 1018. Would you expect 1075 to have four times the hardness? Explain using what you know about how pearlite fraction relates to hardness.

Answer:

Step 3 — Engineering Equations & Concept Check

Lever Rule — Ferrite/Pearlite Fraction (hypoeutectoid steel)

$$\text{Fraction pearlite} \approx (C_0 - 0.022) / (0.76 - 0.022) \quad \text{Fraction ferrite} = 1 - \text{Fraction pearlite}$$

C_0 — carbon content of the steel, wt% · 0.022 — maximum carbon solubility in ferrite at room temperature, wt% · 0.76 — eutectoid composition, wt% C

C1. Using the lever-rule equation, does increasing carbon content from 1018 (~0.18% C) to 1045 (~0.45% C) increase or decrease the pearlite fraction? Does this match published microstructure descriptions for these grades?

Answer:

C2. If a steel's carbon content exceeded 0.76%, the equation above would give a pearlite fraction greater than 100%. What actually happens microstructurally past the eutectoid composition that this equation doesn't capture?

Answer:

MODULE 4 · WEEK 5 | Stainless Steel Families

Stainless steel resists corrosion through a passive chromium oxide layer, but different stainless families reach very different microstructures from that same starting point. 304 and 316 are both austenitic (FCC, non-magnetic) — 316 differs mainly by a molybdenum addition for chloride resistance. 17-4PH is a precipitation-hardening martensitic grade with a completely different strengthening mechanism and microstructure. Researching all three side by side shows that "stainless" describes a corrosion behavior, not a single microstructure.

Alloys of Record: D — 304 austenitic stainless steel · E — 316 austenitic stainless steel · F — 17-4PH precipitation-hardening stainless steel

Research Sources: PREN / Stainless Steel reference page (course), AZoM.com, MatWeb entries for 304, 316, and 17-4PH

Step 1 — Research & Documentation

Composition of 304, 316, and 17-4PH (note the Mo addition in 316 and the Cu/Nb precipitation-hardening chemistry in 17-4PH); PREN for each (or the Cr/Mo/N values to calculate it); typical crystal structure (austenitic FCC vs. martensitic) and why 17-4PH's aging heat treatment differs fundamentally from 304/316's annealed condition.

Step 2 — Short-Answer Questions

Q1. Austenitic stainless steels (D, E) are FCC and typically show prominent annealing twins; the martensitic precipitation-hardening grade (F) does not. Explain why FCC crystal structure favors annealing twin formation while a martensitic structure does not.

Answer:

Q2. 316 adds roughly 2–3% molybdenum compared to 304. That addition doesn't change the crystal structure or general grain appearance, yet 316 is specified over 304 in marine and chloride-heavy environments. What does Mo actually do?

Answer:

Q3. Specimen F's microstructure (martensitic matrix with fine precipitates) is completely different from D and E's austenitic twinned grain structure. Describe that difference, and explain why "stainless" doesn't guarantee a particular microstructure the way "1018" or "1045" does for carbon steel.

Answer:

Q4. 17-4PH is strengthened by precipitation during aging, not by the FCC structure that produces twinning in D and E. Based on your research, would you expect 17-4PH to be more or less prone to work hardening during machining than 304? Explain your reasoning.

Answer:

Step 3 — Engineering Equations & Concept Check

Pitting Resistance Equivalent Number (PREN)

$$\text{PREN} = \% \text{Cr} + 3.3(\% \text{Mo}) + 16(\% \text{N})$$

%Cr, %Mo, %N — weight percent chromium, molybdenum, and nitrogen in the alloy

C1. 316 stainless adds roughly 2–3% Mo compared to 304. Using the PREN formula, estimate how many PREN points that Mo addition is worth, and explain why the formula weights Mo so much more heavily than Cr.

Answer:

C2. 17-4PH is a precipitation-hardening martensitic stainless, not an austenitic grade like 304/316. Is PREN still a meaningful comparison across all three families, or does 17-4PH's different base microstructure change what "corrosion resistance" depends on?

Answer:

MODULE 5 · WEEK 6 | Nonferrous Metals & Phase Diagrams

Nonferrous alloys rely on different hardening mechanisms than steel. Aluminum uses precipitation hardening; brass relies on solid-solution strengthening. Both develop grain structures broadly similar to steel's but with very different twin densities — a direct consequence of stacking fault energy.

Alloys of Record: G — 6061-T6 Aluminum alloy · K — 70/30 Brass (Cu-Zn)

Research Sources: Aluminum Alloy & Temper Designation page, Copper/Brass/Bronze Codes page (course), AZoM.com, MatWeb entries for 6061-T6 and 260 brass, the Aluminum Association's free composition reference ("Pink Sheets," aluminum.org)

Step 1 — Research & Documentation

Composition of 6061-T6 and 70/30 brass; the T6 temper designation (what "solution heat treated and artificially aged" means); typical stacking fault energy comparison between aluminum and copper alloys and how it relates to twinning frequency.

Step 2 — Short-Answer Questions

Q1. Compare the annealing twin density typically reported for Specimen K (70/30 brass) vs. Specimen D (304 stainless, Module 4). Both are FCC — which alloy is known for more prominent twins, and why (stacking fault energy)?

Answer:

Q2. The 6061-T6 designation means solution heat treated and artificially aged. What does aging do to the microstructure, and why does it increase hardness?

Answer:

Q3. 6061-T6 is strengthened by fine Mg_2Si precipitates. Describe their typical size/distribution (per your research) and how they strengthen the alloy.

Answer:

Q4. Using the Al-Mg-Si phase diagram concept, explain why solution treatment must precede aging. What happens if the solution treatment step is skipped?

Answer:

Step 3 — Engineering Equations & Concept Check

Lever Rule (general two-phase form)

$$W_{\alpha} = (C_{\beta} - C_0) / (C_{\beta} - C_{\alpha})$$

W_{α} — weight fraction of phase α at a given temperature · C_0 — overall alloy composition · C_{α} , C_{β} — compositions of phases α and β at the ends of the tie line

C1. The lever rule you used for steel in Module 3 is a special case of this general two-phase equation. What has to be true about a phase diagram for the lever rule to apply at all?

Answer:

C2. Aluminum alloys strengthen by precipitation hardening, and brass strengthens by solid-solution strengthening. Neither mechanism is described by the lever rule alone — what is the lever rule actually telling you (and not telling you) about a two-phase microstructure?

Answer:

MODULE 6 · WEEK 7 | Mechanical Testing

Mechanical tests produce numbers; microstructure explains why. Hardness testing leaves a permanent indent whose geometry encodes the hardness value. Grain size is a quantitative microstructural measurement (ASTM E112) that directly correlates to strength and toughness through the Hall-Petch relationship. The Vickers-indent math you previewed in Module 2 becomes official here — you'll apply the same HV formula (in reverse, from a published reference value) to every new alloy through the rest of the semester, building a running hardness comparison across the whole specimen set.

Alloys of Record: A — 1018 steel (normalized) — reference HV value used for Q1 (Appendix A) · B — 1045 steel (normalized) — reference HV value used for Q1 (Appendix A)

Research Sources: Hardness Testing Scales reference page (course), AZoM.com, a free grain-size technical article (e.g., Struers' "Grain Size Measurement Methods" academy page or Buehler's technical notes) — not the paywalled ASTM E112 standard itself

Step 1 — Research & Documentation

Published/typical ASTM E112 grain size numbers for normalized 1018 and 1045; confirm the HV values you'll use in Q1 against a second source if possible.

Step 2 — Short-Answer Questions

Q1. Specimens A (1018 steel) and B (1045 steel) have published/typical hardness values of about 125 HV and 170 HV respectively (Appendix A). For each, use $HV = 1.854 \times F / d_{avg}^2$ ($F = 1 \text{ kgf}$) to solve for d_{avg} — the indent diagonal that reading corresponds to. Record both the given HV and your calculated d_{avg} as the first two entries in your Hardness Log.

Answer:

Q2. Research a typical/published ASTM E112 grain size number (G) for normalized 1018 steel, and using $N = 2^{(n-1)}$, calculate the corresponding number of grains per square inch at 100× magnification.

Answer:

Q3. A tensile test on 1045 normalized steel gives $UTS \approx 750 \text{ MPa}$. Using the empirical relation $UTS \approx 3.3 \times HV$, does your recorded HV predict a consistent UTS?

Answer:

Q4. Impact toughness (Charpy) is sensitive to grain size. If the grain size of Specimen B (1045 steel) were refined by heat treatment, would toughness increase or decrease — and why?

Answer:

Step 3 — Engineering Equations & Concept Check

Vickers Hardness & Hardness-Strength Correlation

$$HV = 1.854 \times F / d^2 \quad UTS \text{ (MPa)} \approx 3.3 \times HV \text{ (empirical, steels)}$$

F — applied load, kgf; d — mean diagonal length of the indent, mm · UTS — ultimate tensile strength; HV — Vickers hardness number

C1. The Vickers equation puts diagonal length d in the denominator, squared. If two operators measure the same indent but one measures d 10% too large, will their calculated HV be too high or too low — and by roughly what factor?

Answer:

C2. The $UTS \approx 3.3 \times HV$ correlation is empirical, not derived from first principles, and it's only reliable for steels. Why might this same multiplier give a wrong answer if applied to an aluminum alloy or a brass?

Answer:

MODULE 7 · WEEK 8 | NDT Methods

NDT methods detect discontinuities indirectly — through magnetic flux leakage, ultrasonic reflection, or surface-penetrant bleed-out. Metallography is the destructive confirmation method NDT is designed to avoid — a distinction that matters most for parts too expensive or too critical to sacrifice. This module bridges the two approaches.

Alloys of Record: A — 1018 steel (baseline reference) · L — 360 leaded brass (discrete second-phase particles) · P — Ti-6Al-4V titanium (aerospace-grade, NDT-first material)

Research Sources: NDT Methods reference page, Copper/Brass/Bronze Codes page, Titanium Grades page

Step 1 — Research & Documentation

Typical inclusion content/type in rolled 1018 steel; how lead particles are distributed in 360 brass and why they're added intentionally; why Ti-6Al-4V aerospace components are inspected by NDT rather than sectioned routinely.

Step 2 — Short-Answer Questions

Q1. In Specimen L (360 brass), the lead particles are an intentional alloying addition, not a defect — yet under magnification they can resemble discrete dark inclusions, similar to how a real defect might appear. Which NDT method — PT, MT, or UT — would best distinguish an engineered particle from an unwanted inclusion in a real part, and why can't NDT alone always make that distinction?

Answer:

Q2. MnS inclusions in rolled 1018 steel are typically elongated and aligned with the rolling direction. Would MT (magnetic-particle inspection) produce a strong or weak indication on this type of feature in a ferromagnetic part? Explain.

Answer:

Q3. Titanium aerospace components like Specimen P (Ti-6Al-4V titanium) are almost never sectioned for metallography in service — they're inspected by NDT throughout their service life instead. Given what titanium costs and where it's used, explain why destructive metallography is reserved for failure investigations and process qualification, not routine inspection.

Answer:

Q4. Metallography destroys the part. Give two situations where destructive metallographic examination is justified over NDT in production or maintenance.

Answer:

Step 3 — Engineering Equations & Concept Check

Ultrasonic Wavelength & Detection Limit

$$\lambda = v / f \quad d_{\min} \approx \lambda / 2$$

λ — ultrasonic wavelength in the material; v — sound velocity in the material · f — transducer frequency; $d_{\min}$ — smallest defect size reliably detectable

C1. Using $\lambda = v/f$, if an inspector needs to detect a smaller defect, should they choose a higher-frequency or lower-frequency transducer? What trade-off in penetration depth comes with that choice?

Answer:

C2. NDT methods like UT and magnetic-particle inspection are indirect — they infer a defect exists rather than showing it directly the way metallography does. What kinds of internal features could two specimens share in a UT signature but actually be microstructurally very different?

Answer:

MODULE 8 · WEEK 9 | Quality Standards & Defect Recognition

Quality standards require documented, reproducible measurements. ASTM E112 for grain size and ASTM E45 for inclusion rating both require systematic examination with written records — this module practices that discipline using published reference data.

Alloy of Record: A — 1018 steel — formal ASTM E112 grain size report

Research Sources: Quality Standards & Codes reference page (course), free technical articles covering ASTM E112 (grain size) and E45 (inclusion rating) methodology (e.g., Struers' or Buehler's knowledge/technical-note pages) — not the paywalled ASTM standards themselves

Step 1 — Research & Documentation

Typical/published ASTM E112 grain size number for normalized 1018 steel; typical ASTM E45 inclusion types and severity ratings reported for a clean, normalized low-carbon steel.

Step 2 — Short-Answer Questions

Q1. Research a typical/published ASTM E112 grain size number for normalized 1018 steel and record it as G.

Answer:

Q2. Research what ASTM E45 inclusion types (A, B, C, D) are typically reported for a clean, normalized low-carbon steel like 1018, and what severity level (Thin or Heavy) is typical. Present your findings in a table.

Answer:

Q3. Using the grain size number you researched in Q1, does normalized 1018 typically meet a material certificate requirement of "ASTM Grain Size 7 minimum"? (G7 $\approx$ 32 μ m grain diameter.)

Answer:

Q4. A quality inspector must document a metallographic examination per written procedure. List four data items that must appear in a conforming examination record.

Answer:

Step 3 — Engineering Equations & Concept Check

ASTM E112 Grain Size Number

$$N = 2^{(n-1)}$$

N — number of grains per square inch at 100× magnification · n — ASTM grain size number

C1. Using $N = 2^{(n-1)}$, does a higher ASTM grain size number (n) correspond to larger or smaller grains? Reconcile your answer with the Hall-Petch relationship from Module 1 — does a higher n mean a stronger or weaker material?

Answer:

C2. ASTM E112 requires a documented, systematic counting procedure rather than an "eyeball estimate" of grain size. Why does reproducibility matter more in a quality-standards context than it might in a research setting where you're just trying to understand a general trend?

Answer:

MODULE 9 · WEEK 10 | Wrought Aluminum Alloy Systems

Not every aluminum alloy is strengthened the same way. 6061 is an Al-Mg-Si (6xxx) alloy, precipitation-hardened by fine Mg_2Si particles; 2024 is an Al-Cu (2xxx) alloy, precipitation-hardened by a different phase with coarser, more visible constituent particles. Researching both wrought alloys side by side shows how two grades can share a strengthening mechanism — aging — while producing very different microstructures and very different corrosion behavior. This week also restarts the hardness-measurement thread from Module 6 — you'll research and log HV for both G and J, the first entries in a running dataset you'll keep building all semester.

Alloys of Record: G — 6061-T6 aluminum alloy · J — 2024-T3 aluminum alloy

Research Sources: Aluminum Alloy & Temper Designation page (course), AZoM.com, MatWeb entries for 6061-T6 and 2024-T3

Step 1 — Research & Documentation

Composition of 6061 (Al-Mg-Si) and 2024 (Al-Cu); typical constituent-particle chemistry each produces; published corrosion-resistance comparison between the two; why 2024 is often sold clad with pure aluminum.

Step 2 — Short-Answer Questions

Q1. Research the typical constituent-particle size and density in 6061-T6 (Al-Mg-Si) versus 2024-T3 (Al-Cu). Which alloy is known for coarser, more numerous particles, and does that match what you'd expect from Al-Cu vs. Al-Mg-Si intermetallic chemistry?

Answer:

Q2. Both alloys are strengthened by aging, yet 2024 is notably less corrosion-resistant than 6061 in service (it's often sold clad with pure aluminum for exactly this reason). Based on the alloying elements each contains, propose a reason why.

Answer:

Q3. If you swapped the aging treatment of these two alloys — 2024's schedule onto 6061's composition and vice versa — would you expect similar strengthening in both? Why or why not?

Answer:

Q4. 6061 is a general-purpose structural alloy; 2024 is an aerospace skin alloy. Based on your research, suggest one property difference — beyond corrosion resistance — that would drive an engineer to choose one over the other.

Answer:

Q5. Appendix A gives published/typical hardness values for Specimens G (6061-T6 aluminum, ≈ 95 HV) and J (2024-T3 aluminum, ≈ 125 HV). Using the relationship from Module 6 ($HV = 1.854 \times F / d_{avg}^2$), calculate the indent diagonal d_{avg} each value corresponds to, and record both HV values in your Hardness Log. Which alloy reads harder, and does that match the constituent-particle density you compared in Q1?

Answer:

Step 3 — Engineering Equations & Concept Check

Precipitation (Orowan) Strengthening

$$\Delta\sigma \approx Gb / L$$

$\Delta\sigma$ — strength increase from precipitates • G — shear modulus; b — Burgers vector magnitude (a lattice property) • L — average spacing between precipitate particles

C1. Both Hall-Petch (Module 1) and the Orowan equation above describe dislocations being blocked by obstacles — grain boundaries in one case, precipitates in the other. What do d (grain diameter) and L (precipitate spacing) have in common in how they show up in each equation?

Answer:

C2. 6061 and 2024 both strengthen by precipitation hardening but reach different peak strengths. Based on the Orowan equation, what two microstructural features would you expect to differ between a well-aged alloy and a slightly under-aged one?

Answer:

C3. Your G specimen is 6061-T6 (artificially aged to peak strength); your J specimen is 2024-T3 (only naturally aged at room temperature, plus cold work). 2024 ages usefully at room temperature; 6xxx alloys like 6061 need an artificial aging step to reach a useful strength. Given that difference in natural-aging behavior, why does it make sense that 2024 sheet commonly ships as T3/T4 while 6061 commonly ships as T6?

Answer:

MODULE 10 · WEEK 11 | Aluminum Alloys: Heat-Treatable vs. Weldable Structural Grades

Alloy selection is driven by more than peak strength — how an alloy performs after fabrication matters just as much. 6061-T6 reaches its strength through Mg_2Si precipitation, but welding (or anything that heats the joint above the aging temperature) dissolves and coarsens that precipitate structure, softening the heat-affected zone. 5086 is a non-heat-treatable Al-Mg alloy that gets its strength entirely from magnesium in solid solution — with no precipitate structure to disrupt, it keeps its strength through a weld thermal cycle far better than 6061-T6 does, which is why welded aluminum structures (boat hulls, tanks, weldments) lean on 5xxx alloys almost exclusively. You'll revisit G and research your first HV reading on I, adding a new point to the hardness dataset you started in Modules 6 and 9.

Alloys of Record: G — 6061-T6 aluminum alloy (revisit, Modules 5 & 9) · I — 5086 aluminum alloy

Research Sources: Aluminum Alloy & Temper Designation page (course), AZoM.com, MatWeb entries for 6061-T6 and 5086

Step 1 — Research & Documentation

Confirm 5086's composition (Al-Mg, non-heat-treatable) and its work-hardening (strain-hardening) behavior; research published guidance on why 5xxx alloys are preferred for welded marine aluminum structures over 6xxx.

Step 2 — Short-Answer Questions

Q1. Describe, based on your research, the presence or absence of a precipitate phase in each specimen (Mg_2Si in G vs. none in I) that reflects the heat-treatable vs. non-heat-treatable difference between them.

Answer:

Q2. Based on your research on 6061-T6's Mg_2Si precipitate structure, what would you expect a welded 6061-T6 joint's heat-affected zone to look like microstructurally compared to the base metal?

Answer:

Q3. If a sheet of 5086 were cold-formed, what microstructural change (grain elongation, dislocation density) would you expect compared to the as-received condition?

Answer:

Q4. A designer needs a welded aluminum bracket for a marine application. Based on the strengthening mechanism of each alloy, which would you recommend — 6061-T6 or 5086 — and why?

Answer:

Q5. Appendix A gives published/typical hardness values for Specimens G (6061-T6 aluminum, ≈95 HV) and I (5086 aluminum, ≈70 HV). Using the Module 6 relationship, calculate the indent diagonal each corresponds to, and record both in your Hardness Log. Does this data suggest 5086 comes close to 6061-T6's strength, or does the precipitation-hardened specimen pull well ahead?

Answer:

Step 3 — Engineering Equations & Concept Check

Hollomon Strain-Hardening Equation

$$\sigma = K \cdot \epsilon^n$$

σ — true stress; K — strength coefficient · ϵ — true plastic strain; n — strain-hardening exponent (a formability indicator)

C1. A higher strain-hardening exponent n means a metal keeps getting stronger as it's deformed, rather than strength leveling off quickly. Why would a cold-formed application like 5086 favor a higher n value over a lower one?

Answer:

C2. 6061-T6 is chosen for its high as-delivered strength, not for how well it retains that strength after welding. What does that tell you about why "high-strength" and "weld-ready" can be two different selection criteria for the same structural application?

Answer:

MODULE 11 · WEEK 12 | Machining Processes

Machining removes material by chip formation, but the cutting zone generates heat and stress that can alter near-surface microstructure. Surface finish and subsurface integrity are quality-critical: a ground surface that looks perfect to the eye may carry thermal or mechanical damage detectable only under a microscope. This week also puts your Hardness Log to work: the HV values you've already recorded for A, B, and G turn out to predict a lot about how each one machines.

Alloys of Record: A — 1018 steel (baseline machinability reference) · B — 1045 steel · G — 6061 aluminum (machinability comparison)

Research Sources: AZoM.com and MatWeb machinability ratings for 1018, 1045, and 6061, manufacturer machining data sheets (freely downloadable, no login required)

Step 1 — Research & Documentation

Research what near-surface microstructural changes machining typically causes in carbon steel and aluminum (a plastically deformed layer, grain distortion, possible white-layer formation in steel); published machinability ratings for 1018, 1045, and 6061.

Step 2 — Short-Answer Questions

Q1. Research what near-surface microstructural changes are typically caused by machining in a carbon steel like 1018 (e.g., a plastically deformed layer, grain distortion near the surface). Describe what you'd expect to find and how it would differ from the bulk interior structure.

Answer:

Q2. In machining, tool wear increases cutting forces and heat. What microstructural change could this cause in a high-carbon steel workpiece, and how would it appear under magnification?

Answer:

Q3. Surface finish affects fatigue life. Based on your research, identify one near-surface microstructural feature (in any of these specimens) that would act as a fatigue crack initiation site on a machined surface in service.

Answer:

Q4. Research which typically shows more machining-induced near-surface deformation — steel or aluminum — and explain why aluminum's lower strength/stiffness might make it more susceptible.

Answer:

Q5. Pull your recorded HV values for Specimens A (1018), B (1045), and G (6061) from your Hardness Log. Rank them softest to hardest, and explain how that ranking predicts which would need the lowest cutting speed and which would show the most tool wear in machining.

Answer:

Step 3 — Engineering Equations & Concept Check

Taylor Tool-Life Equation

$$V \cdot T^n = C$$

V — cutting speed; T — tool life (time to failure) · n — Taylor exponent (tool-material dependent); C — constant (workpiece/tool-material dependent)

C1. Using $V \cdot T^n = C$, if cutting speed V is increased, does tool life T go up or down? Now connect this to your Hardness Log: would you expect a harder specimen (higher HV) to need a lower or higher cutting speed to hit the same tool life?

Answer:

C2. The Taylor equation only accounts for cutting speed — it says nothing about microstructure directly. Based on what you researched in Modules 9–10, what microstructural feature (not captured by V or T) could still cause two specimens with the same HV to machine very differently?

Answer:

MODULE 12 · WEEK 13 | Copper Alloys: Strengthening Mechanisms

Powder metallurgy and additive manufacturing are covered in this week's lecture material; this module's research thread instead continues the strengthening-mechanism theme using pure and alloyed copper. Specimen N (110 copper) is commercially pure copper — no alloying, no precipitates, just grain boundaries and annealing twins. Specimen O (182 chromium copper) is a precipitation-hardening alloy that trades a little electrical conductivity for a real strength increase — the same aging mechanism you've now researched in steel (17-4PH) and aluminum (2024), applied to a third base metal entirely. You'll also research HV readings for N and O, extending the semester's hardness dataset into a third base metal.

Alloys of Record: N — 110 copper (commercially pure) · O — 182 chromium copper (precipitation-hardening)

Research Sources: Copper/Brass/Bronze Codes page (course), Copper Development Association alloy pages (alloys.copper.org) for C11000 and C18200, MatWeb entries for both

Step 1 — Research & Documentation

Composition and conductivity ratings for 110 copper and 182 chromium copper; the strengthening precipitate in chromium copper; why alloying reduces electrical conductivity generally.

Step 2 — Short-Answer Questions

Q1. Research the typical grain/twin structure of pure copper (N) versus chromium copper (O). Which is "simpler," and why would you expect that from a single-element metal versus an alloyed one?

Answer:

Q2. You've now researched precipitation hardening in three different base metals this semester: 17-4PH stainless steel, 2024-T3 aluminum, and Specimen O (182 chromium copper). List the strengthening phase in each, and state what all three approaches have in common at the mechanism level.

Answer:

Q3. Pure copper (Specimen N) has excellent electrical conductivity; the chromium addition in Specimen O reduces that conductivity somewhat. Why does adding any alloying element — even in small amounts — generally reduce electrical conductivity in a metal?

Answer:

Q4. Powder metallurgy (this week's lecture topic) sinters metal powder into a part without ever fully melting it. Based on the wrought, fully-dense alloys you've researched all semester, what microstructural feature would you expect to see in a PM part that none of them would show?

Answer:

Q5. Appendix A gives published/typical hardness values for Specimens N (110 copper, ≈ 50 HV) and O (182 chromium copper, ≈ 130 HV). Using the Module 6 relationship, calculate the indent diagonal each corresponds to, and record both in your Hardness Log. Given the precipitation-hardening thread from Q2, how large is the jump from N to O — and does its size look more like the jump you'd expect from steel (17-4PH) or from aluminum (2024)?

Answer:

Step 3 — Engineering Equations & Concept Check

Matthiessen's Rule (Electrical Resistivity)

$$\rho_{\text{total}} = \rho_{\text{thermal}} + \rho_{\text{impurity}}$$

ρ_{total} — total electrical resistivity · ρ_{thermal} — resistivity from thermal lattice vibration (temperature-dependent) · ρ_{impurity} — resistivity from solute atoms, precipitates, and defects (composition-dependent)

C1. Pure copper (Specimen N) has no alloying additions, while chromium copper (Specimen O) does. Using Matthiessen's rule, explain why O has higher electrical resistivity than N even at the same test temperature.

Answer:

C2. The same precipitates that raise chromium copper's strength (via the Orowan mechanism from Module 9) also raise its resistivity (via Matthiessen's rule here). Why can't you get the strength benefit of precipitation hardening in copper "for free," with no trade-off?

Answer:

MODULE 13 · WEEK 14 | Brass Alloys: Composition and Microstructure

Welding process fundamentals are covered in this week's lecture material; this module's research thread instead examines how zinc content changes brass microstructure across three grades. Specimens K (cartridge brass, 70/30 Cu-Zn, revisit from Module 5) and L (free-machining brass, revisit from Module 7) return for a direct side-by-side comparison, joined by Specimen M (naval brass), researched here for the first time. K is single-phase alpha; L adds a small lead addition for machinability; M crosses into a two-phase alpha-beta structure at higher zinc content. Together they show a single compositional variable — zinc — driving a real change in phase count, not just grain size. None of K, L, or M has a logged hardness value yet, so this week you'll research the first HV reading for all three and add them to your running dataset.

Alloys of Record: K — 260 brass (cartridge, single-phase alpha; revisit, Module 5) · L — 360 brass (lead, free-machining; revisit, Module 7) · M — 464 brass (naval, duplex alpha-beta)

Research Sources: Copper/Brass/Bronze Codes page (course), Copper Development Association alloy pages (alloys.copper.org) for C26000, C36000, and C46400, MatWeb entries for the same, a freely available Cu-Zn phase diagram (e.g., a materials-science course page or Wikipedia)

Step 1 — Research & Documentation

Zinc content for 260, 360, and 464 brass; where the alpha/beta phase boundary falls on the Cu-Zn phase diagram; the role of lead in 360's machinability; published hardness for all three (Appendix A).

Step 2 — Short-Answer Questions

Q1. Research whether annealing twins remain visible in leaded brass (L) and duplex alpha-beta brass (M) the way they are in single-phase alpha brass (K), or whether the added lead/beta phase makes them harder to distinguish.

Answer:

Q2. Lead is present in Specimen L (360 brass) as discrete, essentially insoluble particles in the copper-zinc matrix. Explain how that insolubility is exactly what makes it useful for machinability.

Answer:

Q3. Specimen M (464 brass) is the only one of the three with two distinct phases (alpha and beta) rather than one. Based on the zinc contents of these three brasses, what does that tell you about where the alpha/beta phase boundary falls on the Cu-Zn phase diagram?

Answer:

Q4. A component needs to be both machined economically and resistant to dezincification in seawater. Based on your research into Specimens K, L, and M, would you specify 360 or a duplex naval brass — and what's the tradeoff?

Answer:

Q5. Appendix A gives published/typical hardness values for Specimens K (260 brass, ≈ 85 HV), L (360 brass, ≈ 80 HV), and M (464 brass, ≈ 90 HV). Using the Module 6 relationship, calculate the indent diagonal each corresponds to, and record all three in your Hardness Log — their first entries. Does the duplex alpha-beta structure of M read harder than the single-phase alpha of K and L, consistent with what phase count alone would predict?

Answer:

Step 3 — Engineering Equations & Concept Check

Applying the Lever Rule to Cu-Zn Brass

$$W_{\beta} = (C_0 - C_{\alpha}) / (C_{\beta} - C_{\alpha})$$

W_{β} — weight fraction of beta phase · C_0 — overall zinc content of the brass · C_{α} , C_{β} — zinc content at the α and β phase-boundary compositions (from the Cu-Zn phase diagram)

C1. K (cartridge brass, $\sim 30\%$ Zn) is single-phase alpha, while M (naval brass, higher Zn) is two-phase alpha-beta. Using the lever-rule logic above, what has to be true about M's zinc content relative to the alpha-phase solubility limit for a second phase to appear at all?

Answer:

C2. L (free-machining brass) gets its machinability from a small lead addition, not from a phase change. Does lead show up anywhere in the Cu-Zn lever-rule calculation above? What does that tell you about the limits of a two-component phase diagram when a real alloy has three or more elements?

Answer:

MODULE 14 · WEEK 15 | Precipitation Hardening: A Cross-Family Comparison

Weld metallurgy and failure analysis are covered in this week's lecture material. This module instead revisits three specimens researched earlier in the semester — Specimen F (17-4PH stainless steel, Module 4), Specimen J (2024-T3 aluminum, Module 9), and Specimen O (182 chromium copper, Module 12) — to directly compare how the same strengthening mechanism, precipitation hardening, plays out across three different base metals, crystal structures, and precipitate chemistries. You'll research your first HV reading on F here, then pull your already-logged values for J and O from your Hardness Log — turning three separate weekly measurements into a single cross-family strength comparison.

Alloys of Record: F — 17-4PH stainless steel (revisit, Module 4) · J — 2024-T3 aluminum (revisit, Module 9) · O — 182 chromium copper (revisit, Module 12)

Research Sources: Your Modules 4, 9, and 12 research notes; AZoM.com and MatWeb entries for 17-4PH, 2024-T3, and 182 chromium copper; free precipitation-hardening overview articles (e.g., AZoM.com or a materials-science course page)

Step 1 — Research & Documentation

For each of F, J, and O: base metal, strengthening precipitate phase, and typical precipitate scale (fine/hard-to-resolve vs. coarse/easily visible); base crystal structure (martensitic vs. FCC) for each.

Step 2 — Short-Answer Questions

Q1. Fill in a short table: for each of F, J, and O, name the base metal, the strengthening precipitate phase (research this), and whether that precipitate is typically fine/hard-to-resolve or coarse/easily visible under a microscope.

Answer:

Q2. All three of these alloys go through the same basic aging heat-treatment sequence conceptually: solutionize, quench, age. Explain in your own words what each of those three steps accomplishes, using one of the three specimens as your example.

Answer:

Q3. Precipitation hardening increases strength without changing the base crystal structure (martensitic in F, FCC in J and O). Why is a fine, closely-spaced precipitate more effective at blocking dislocation motion than a coarse, widely-spaced one?

Answer:

Q4. If all three specimens were left at room temperature for years ("natural aging") instead of artificially aged, would you expect them to reach the same final strength as the artificially aged condition? Explain why or why not, using what you know about diffusion rates at room temperature.

Answer:

Q5. Appendix A gives a published/typical hardness value for Specimen F (17-4PH stainless steel, ≈ 375 HV) — its first entry in your Hardness Log. Using the Module 6 relationship, calculate the indent diagonal this value corresponds to. Then pull your already-logged values for J (2024-T3 aluminum) and O (182 chromium copper). Using rough annealed baselines (304 stainless ≈ 150 HV, 6061 ≈ 30 HV, pure copper ≈ 40 HV), rank the three families by relative strength increase from precipitation hardening. Which family gained the most?

Answer:

Step 3 — Engineering Equations & Concept Check

Revisiting Orowan Strengthening Across Three Base Metals

$$\Delta\sigma \approx Gb / L \text{ (Module 9)}$$

G — shear modulus (different for each base metal — steel, aluminum, copper) · b — Burgers vector; L — interparticle spacing

C1. F (17-4PH steel), J (2024 Al), and O (182 Cu) all strengthen by the same Orowan mechanism, but steel's shear modulus G is roughly 3× aluminum's and copper's. Does that alone guarantee 17-4PH will show the largest $\Delta\sigma$ of the three? What else would you need to know about L to answer with confidence?

Answer:

C2. You now have logged HV values for F, J, and O in your Hardness Log. Do the relative HV rankings match what the Orowan equation alone would predict from G , or do differences in precipitate spacing (L) between the three alloys appear to dominate instead?

Answer:

MODULE 15 · WEEK 16 | Course Capstone & Synthesis

Corrosion mechanisms and protection methods are covered in this week's lecture material and the course reference pages — including a direct callback to Specimens D and E (304 vs. 316) from Module 4, since molybdenum content and PREN are exactly why 316 outperforms 304 in chloride environments. This final module is a synthesis exercise: no new alloy, just your own research notes from across the semester, organized into one final comparison. Your Hardness Log — built from Modules 6, 9, 10, 12, 13, and 14 — also comes back into play as evidence for the Unit 3 link in this week's synthesis question.

Alloys of Record: No new alloy — synthesis exercise using research notes from Modules 1–14

Research Sources: Your own portfolio notes and answered questions from Modules 1–14; PREN / Corrosion Protection reference pages

Using Your Research for This Module

1. Gather your research notes, tables, and answered questions from Modules 1–14.
2. Re-read your notes on Specimens D and E (304 vs. 316, Module 4) before answering the PREN-related question.
3. Re-read your notes on Specimens F (17-4PH stainless steel), J (2024-T3 aluminum), and O (182 chromium copper) (Modules 4, 9, 12) before answering the precipitation-hardening synthesis question.
4. No new research is required for this module — it is a written synthesis exercise using your notes from Modules 1–14.
5. Review the full course arc (Units 1–5) before writing your synthesis answer.

Step 2 — Short-Answer Questions

Q1. Across every alloy you researched this semester, identify one example each of a BCC structure, an FCC structure, and an HCP structure (Specimen P, Ti-6Al-4V titanium, Module 7). List the specimen and family for each.

Answer:

Q2. Annealing twins commonly appear in FCC alloys (304, 316, brass) but not in the BCC carbon steels. Explain why, in terms of stacking fault energy and crystal structure, twinning is common in some metals and essentially absent in others.

Answer:

Q3. You also researched three strengthening mechanisms this semester: solid-solution strengthening (5086, pure copper — no precipitates), precipitation/age hardening (17-4PH, 2024-T3, chromium copper), and grain-boundary strengthening via Hall-Petch (present to some degree in every specimen). For any one specimen, identify which mechanism(s) applied and explain your reasoning.

Answer:

Q4. Connect the course's five units using your own research as evidence: How does atomic structure/bonding (Unit 1) → alloy family and composition (Unit 2) → mechanical testing and inspection (Unit 3) → manufacturing process and alloy selection (Unit 4) → service behavior including corrosion resistance (Unit 5) form a single chain? Cite at least three specimens by letter as evidence.

Answer:

Q5. Compile your full Hardness Log (A, B, G, I, J, K, L, M, N, O, F) into one ranked list from softest to hardest. Pick any two adjacent alloys from different families and explain, using composition and microstructure, why their hardness values ended up so close despite being different metals.

Answer:

Step 3 — Engineering Equations & Concept Check

Semester Equation Toolkit (recap)

Hall-Petch: $\sigma_y = \sigma_0 + k_y/\sqrt{d}$ | PREN = %Cr + 3.3(%Mo) + 16(%N) | HV = $1.854F/d^2$, UTS $\approx 3.3 \times HV$ | Orowan: $\Delta\sigma \approx Gb/L$

Definitions as introduced in Modules 1, 4, 6, and 9 — see those pages if you need a refresher.

C1. Hall-Petch and Orowan strengthening both block dislocations, but PREN doesn't describe strength at all — it describes corrosion resistance. Using your Module 4 specimens (D vs. E, 304 vs. 316), explain why a metal can be re-engineered to resist corrosion without changing its strengthening mechanism at all.

Answer:

C2. Across all 15 modules, which single equation from the toolkit above connects most directly to a microstructural feature you could actually observe under a microscope, and which is hardest to connect to a visual observation? Defend your answer using at least two specimens from your semester.

Answer:

Appendix A — Reference Data Directory & Hardness Values

This portfolio is a research deliverable: every alloy below is documented using real published data rather than a live microscope observation. Where a source doesn't carry data for the exact commercial grade/temper used in the physical kit (e.g., 17-4PH stainless, Ti-6Al-4V), use the closest representative published data for the same alloy family and note the substitution in your answer — that's expected, not an error.

Specimen Summary

A — 1018 steel (normalized) · B — 1045 steel (normalized) · C — 1075 steel (normalized) · D — 304 austenitic stainless steel · E — 316 austenitic stainless steel · F — 17-4PH precipitation-hardening stainless · G — 6061-T6 aluminum alloy · I — 5086 aluminum alloy · J — 2024-T3 aluminum alloy · K — 260 brass (cartridge, 70/30) · L — 360 brass (leaded, free-machining) · M — 464 brass (naval, duplex alpha-beta) · N — 110 copper (commercially pure) · O — 182 chromium copper (precipitation-hardening) · P — Titanium Grade 5 (Ti-6Al-4V)

The full specimen kit, including etchants used for each alloy, is also used for in-class demonstrations — see the EXAMET-5 Microscope & Etchant Reference page.

Your Hardness Log (Reference Values + Fill-In Table)

The Reference HV column below is approximate literature/typical data, provided so every hardness-comparison and HV-formula question can still be completed with a consistent dataset — real hardness varies with exact heat treatment, prior cold work, and test method, so feel free to cite a published value you find that differs, noting the discrepancy. Every module that asks you to calculate a d_{avg} or "record this in your Hardness Log" (Modules 6, 9, 10, 12, 13, 14) is asking you to fill in that specimen's row here — by the time you reach Module 15's synthesis question, this table should be a complete, filled-in Hardness Log for Specimens A, B, F, G, I, J, K, L, M, N, and O. Specimens C, D, E, and P are not logged this semester; their reference values are included for completeness only.

Specimen	Alloy / Grade	Module First Logged	Reference HV (Published/Typical)	Your Calculated d_{avg} (mm)	Source You Cited
A	1018 steel (normalized)	Module 6 (Wk 7)	125		
B	1045 steel (normalized)	Module 6 (Wk 7)	170		
C	1075 steel (normalized)	Not logged this semester	220		
D	304 austenitic stainless steel	Not logged this semester	150		
E	316 austenitic stainless steel	Not logged this semester	160		
F	17-4PH precipitation-hardening stainless	Module 14 (Wk 15)	375		
G	6061-T6 aluminum alloy	Module 9 (Wk 10)	95		
I	5086 aluminum alloy	Module 10 (Wk 11)	70		

Specimen	Alloy / Grade	Module First Logged	Reference HV (Published/Typical)	Your Calculated d_avg (mm)	Source You Cited
J	2024-T3 aluminum alloy	Module 9 (Wk 10)	125		
K	260 brass (cartridge, 70/30)	Module 13 (Wk 14)	85		
L	360 brass (leaded, free-machining)	Module 13 (Wk 14)	80		
M	464 brass (naval, duplex alpha-beta)	Module 13 (Wk 14)	90		
N	110 copper (commercially pure)	Module 12 (Wk 13)	50		
O	182 chromium copper (precipitation-hardening)	Module 12 (Wk 13)	130		
P	Titanium Grade 5 (Ti-6Al-4V)	Not logged this semester	340		

Grading Summary

Each module is graded on four components: research documentation completeness, short-answer accuracy, critical-thinking reasoning, and source citation. The 15 weekly modules together are worth 75 points; the holistic end-of-semester portfolio review is worth 25 points — totaling 100 points for the semester.

Research documentation — 1 pt — Credit for each data point researched and properly recorded

Short-answer questions (Step 2) — 2 pts — Credit apportioned across all questions in the set

Critical-thinking questions (Step 3) — 1 pt — Equation applied correctly and reasoning tied to real published data

Source citation — 1 pt — Publication/database name and specific value or table referenced, recorded for each data point used

Total per module — 5 pts

15 modules total — 75 pts — $5 \text{ pts} \times 15 \text{ modules}$

End-of-semester portfolio review — 25 pts — Holistic; all 15 modules as a complete document

TOTAL — 100 pts